MP6551 Motor Drive Characterization

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Abstract

In the **3.2 version of the micro-MMC**, we observed few issues related to the motor drive. First, a **timing offset** has been observed between the MCU-generated PWMs and the motor drive generated PWMs. This internal delay within the motor drive inaccurate timing between the assumed and actual switching times. And secondly, **low MCU-generated PWMs (5<%) are not duplicated** by the Motor drive. Both parameters are important for efficient control hence the characterization of this component. This application note characterizes these parameters and explains how works this part of the micro-MMC .

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1. Description of the issues

1.1 Delay propagation issue.

The microcontroller (MCU) generates PWM signals (PWM1 and PWM2) to drive an external motor drive unit (Monolithic power 's MP6551). This component provides two current feedback signals, which are sampled via the MCU's ADC. Accurate current measurement is achieved by synchronizing ADC triggering with the middle of the low state of the PWM cycle, where the current ripple is minimized.

Unfortunately, a timing offset has been observed between the MCU-generated PWMs and the Motor drive generated PWMs. This internal delay within the Motor drive leads inaccurate timing between the assumed and actual switching times.

As a result:

- The ADC is triggered based on the MCU's PWMs timing and thus does not consider the propagation delay.
- Due to this delay, actual current transitions occur later than expected.

This causes the ADC to sample current during unexpected moments leading to inaccurate measurement.

1.2 Low Duty cycle input issue

The MCU could, depending on the algorithm, generates PWMs with low duty cycle, but it seems that the MP6551 doesn't consider low duty cycle probably due to its Control logic. It is then important to fully understand the valid Duty cycle range valid.

2. Measurement Protocol

To evaluate both duty cycle input range and the propagation delay, we proceed multiple tests on each arm.

- **Test 1 (Sensibility)** : Evaluate the sensibility of the motor drive in function of the duty cycle at a fixed frequency.
- **Test 2 (Delay vs Frequency):** Evaluate the propagation delay in function of the frequency for a 50 % duty cycle.
- **Test 3 (Delay vs Duty Cycle):** evaluate the propagation delay at a fixed frequency in function of the duty cycle.

2.1 Test 1 – Input vs Output Duty Cycle (Sensitivity)

Objective

The purpose of this test is to evaluate the MP6551 motor driver's response to varying input PWM duty cycles at a fixed frequency. Specifically, we aim to determine the minimum effective input duty cycle and to characterize the linearity of the input-to-output PWM reproduction.

Methodology

- A waveform generator (EDU33212A) produces a **100 kHz PWM** signal.
- The duty cycle is incrementally increased from 1% to 98% in 0.1% steps.
- Only one arm of the H-bridge is enabled; the other is disabled.
- Oscilloscope probes are connected to the driver outputs (OUT1 or OUT2).
- The input PWM and the corresponding output signals are recorded for analysis

Results and observations

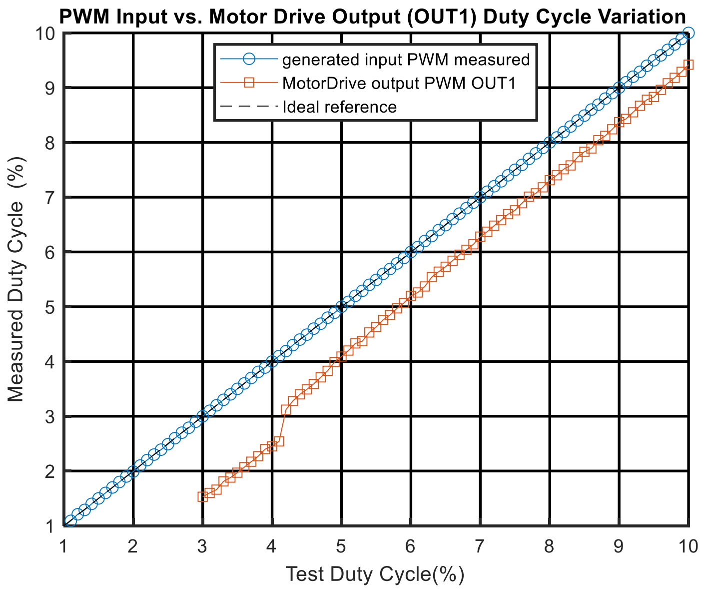


Figure 1 : Input vs Output Duty Cycle – Zoom View (OUT1)

As seen in Figure 1 , the MP6551 output begins reproducing the input signal only once the input duty cycle exceeds approximately 3%. Below this threshold, the device does not generate any output. This suggests an internal minimum pulse width requirement of around around 300ns (at 100 kHz), below which the MP6551 likely filters or ignores incoming signals. It possibly due to the filters by internal timing logic or considered too short to safely drive output FETs.

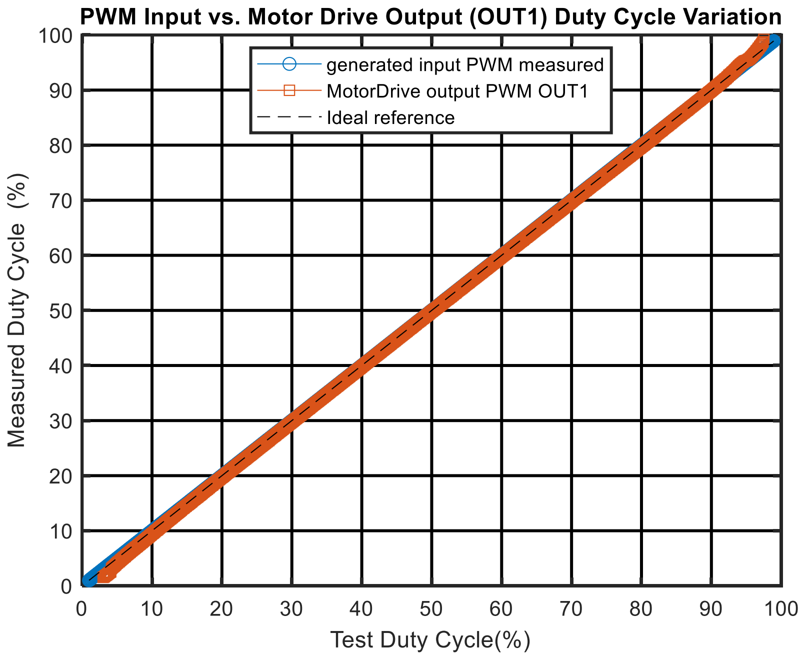
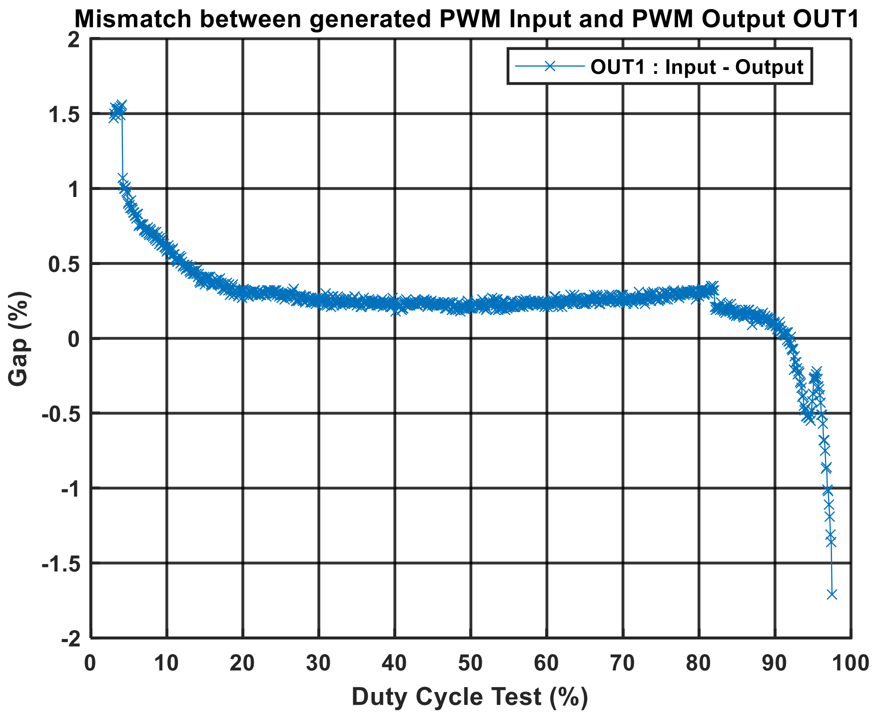


Figure 2 : Input vs Output Duty Cycle – Full range

On the Figure 2, from approximately 4% to 95%, the MP6551 seems to reproduces the input the PWM signal. Except some clear deviation at the low end (<10%) and the high end (>95%), the driver demonstrates a linear and stable response across the majority of the input range.

The following **Erreur ! Source du renvoi introuvable.**, shows clearly the mismatch between the



input and output.

- **The Low Duty Cycle Region** (3%-10%) :
    - Mismatch is strongly positive between 0.5% and 1.5%.

  That indicates that for very low duty cycle, the MP6551 produces an output with lower duty than the input.

- **The Midrange Region** (15 – 80%):
    - Mismatches stabilize around 0.2%-0.3%, showing high linearity

  This is the optimal region for operation indicating that the motor drive's reproduction logic works best here

- The **High Duty Cycle Region (90% – 100%)**
    - Mismatch becomes **strongly negative**, reaching nearly **−1.6%**.
    - At ~95% input, the output fails to track linearly → output duty is **less than expected**.
      Indicating output compression

This behaviour reinforces the importance of avoiding extreme duty cycles for applications requiring accurate torque or current regulation.

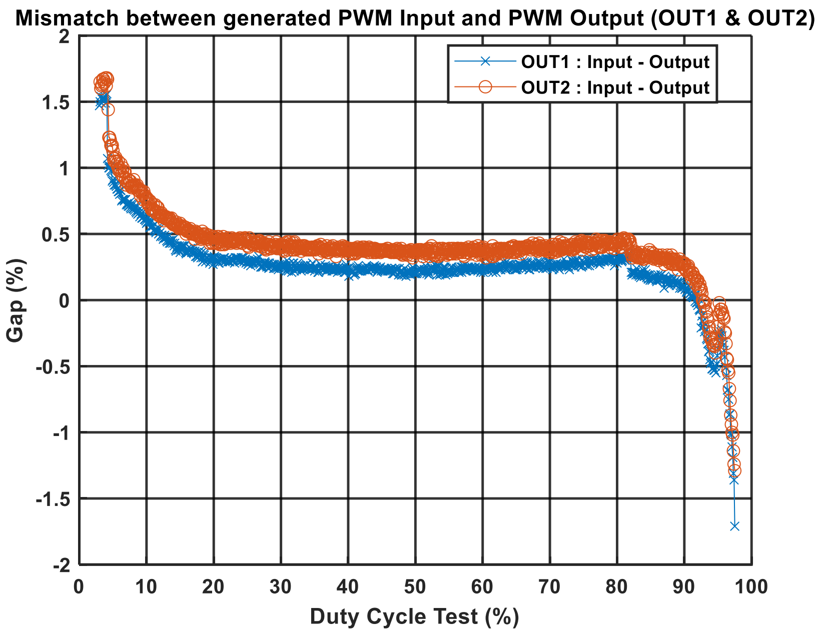


Figure 3 : PWM Duty Cycle Mismatch – OUT1 and OUT2

Figure 3 illustrates the mismatch between the input PWM duty cycle and the output duty cycle for both OUT1 and OUT2. The **evolution of mismatch is highly similar across both channels**, confirming the symmetric behaviour of the MP6551's driver stages. Only **minor deviations**, generally within **±10 ns**, are observed between OUT1 and OUT2, which are negligible for most practical control scenarios.

Conclusion – Test 1 : Input vs Output Duty Cycle

- The **minimum effective input duty cycle** is approximately **3%** at a switching frequency of **100 kHz**; below this threshold, the MP6551 fails to generate output.
- The **optimal linear operating range** lies between **15% and 80% duty cycle**, where the output accurately follows the input with minimal mismatch.
- Both **OUT1 and OUT2 channels** demonstrate **consistent and symmetrical behavior**, making them suitable for **balanced full-bridge operation** within the recommended range.

2.2 Test 2 – Propagation Delay vs Frequency

Objective

This test aims to characterize the **propagation delay** introduced by the MP6551 motor driver as a function of the input PWM **frequency**, while keeping the duty cycle constant. Understanding this behavior is critical for aligning ADC sampling with actual current transitions.

Methodology

- A **waveform generator** (EDU33212A) generates PWM signals with a **fixed duty cycle of 50%**.
- The **PWM frequency is swept** across a range (e.g., from 1of kHz to 150 kHz).
- Only **one arm of the H-bridge is enabled**, and the other is disabled.
- An **oscilloscope probe** is connected to the output of the MP6551 (e.g., OUT1 or OUT2).
- The **time delay between the rising edges** of the input PWM and the output PWM is measured

Results and observations

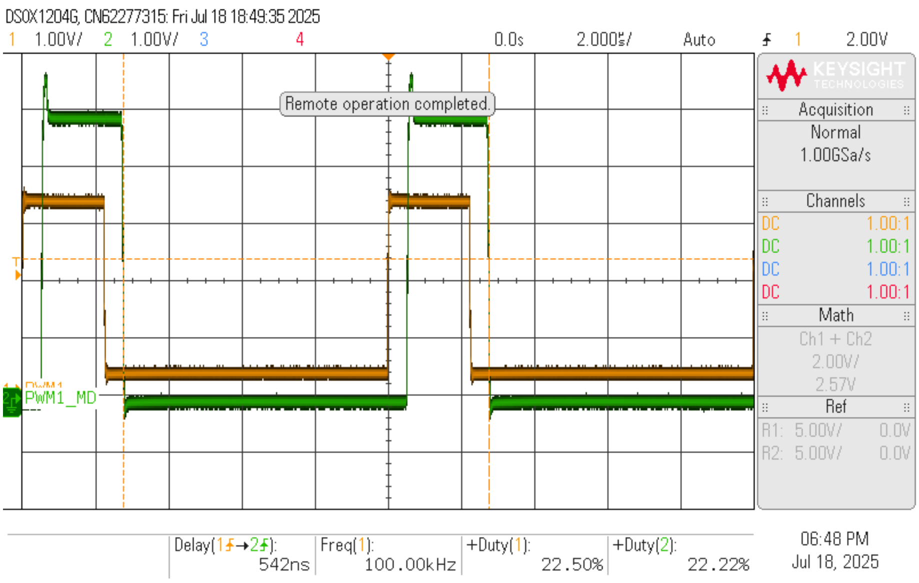


Figure 4 : Input vs Output PWM – 100 kHz (Yellow: Input, Green: Output)

On the Figure 4, we clearly observe the delay between the generated PWM input and the Motor drive-PWM output.

Sur la figure ci-dessous, we observe absurd data due to the fact that we measuring a delay in nanosecond from a 1 Khz frequency

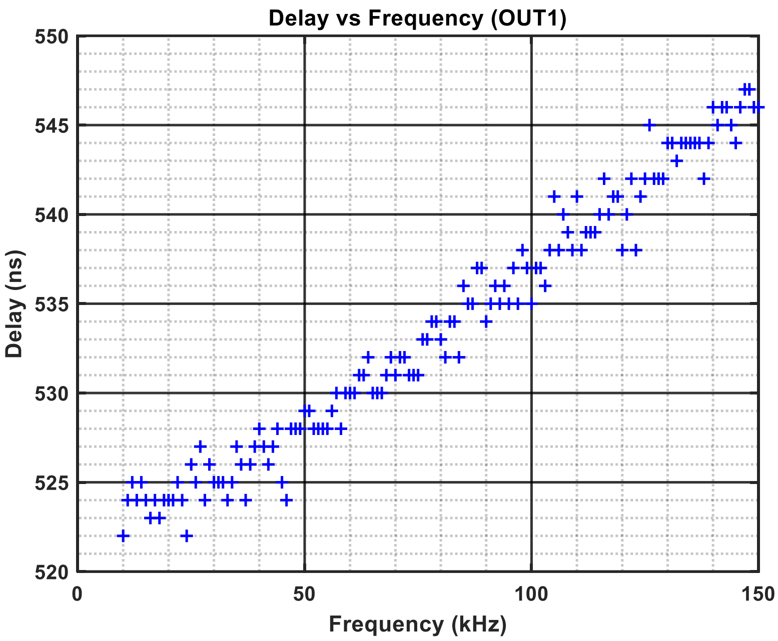


Figure 5 : Propagation Delay vs Frequency – OUT1

Commenté [KT1]: Remove absurd data and start the plot from 10 kHz

We see on Figure 5, the delay for OUT1 remains relatively **stable** across the frequency range from **10 kHz to 150 kHz**, with only a **slight upward trend**.

This trend suggests that the dominant component of delay is from **fixed internal gate logic**, with minimal influence from signal rise/fall time variations across the tested frequency span.

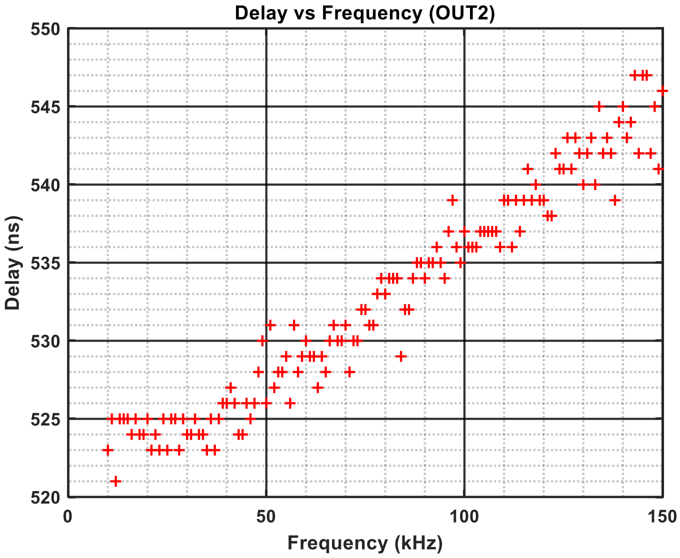


Figure 6 : Propagation Delay vs Frequency - OUT2

On the Figure 6, we observe the delay behaviour of OUT2 **mirrors** that of OUT1, showing a nearly identical curve. This validates that both driver channels are governed by **symmetric internal logic paths**.

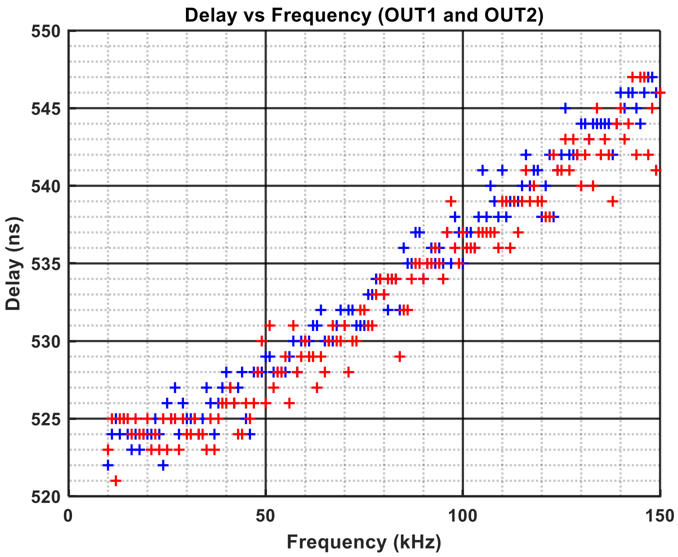


Figure 7 : Delay in Function of the Frequency – OUT1 and OUT2

The **Erreur ! Source du renvoi introuvable.** shows both OUT1 curve and OUT2 curve. When p lotted together, the delay curves of OUT1 and OUT2 confirm **channel matching**, with variations between the two staying within approximately **±5 ns** across the entire frequency range. This behavior is favorable for applications using both sides of the H-bridge in synchronous operation.

Conclusion – Test 2: Delay vs Frequency

- The MP6551 introduces a **consistent and predictable propagation delay** of approximately **535 ns** for both OUT1 and OUT2 at standard operating frequencies.
- The delay remains **stable from 10 kHz to 150 kHz**, enabling reliable **firmware-based compensation** for ADC synchronization.
- Below 10 kHz, measurement error increases due to low-frequency limitations, but this is outside the recommended PWM operating range.
- Both outputs exhibit **symmetric delay characteristics**, simplifying system-level timing alignment in full-bridge applications.

2.3 Test 3 – Propagation Delay vs Duty Cycle

Objective

This test evaluates how the **propagation delay** introduced by the MP6551 varies **with PWM duty cycle** at a fixed frequency. Understanding this dependency is critical for maintaining precise timing alignment in motor control applications, particularly when using ADCs synchronized to switching events

Methodology

For this test, the following measurement protocol has been implemented:

- A PWM signal is generated at a **fixed frequency of 100 kHz** using the EDU33212A waveform generator.
- The **duty cycle is swept** from low to high values (e.g., 1% to 98%).
- Only one half-bridge arm is enabled during each test.
- Oscilloscope probes are placed on the MP6551 outputs (OUT1 and OUT2).
- The **delay** between the rising edge of the input PWM and the corresponding rising edge of the output PWM is recorded.

Results and observations

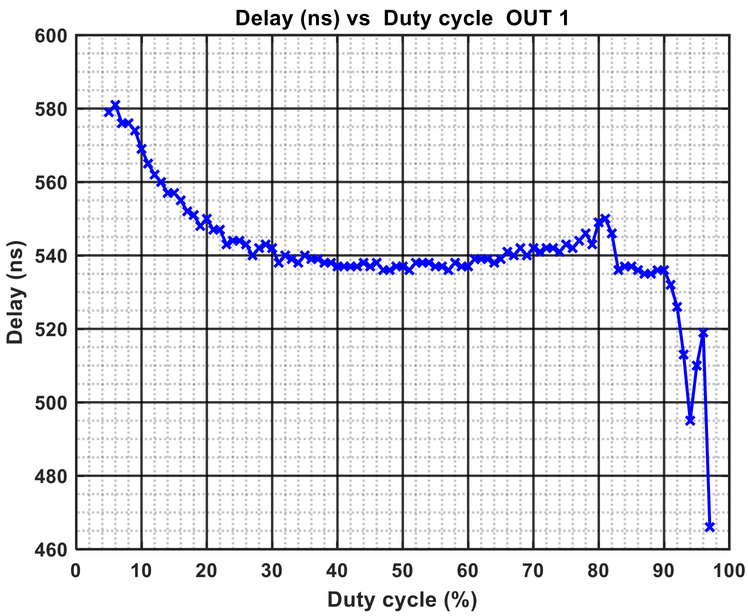


Figure 8 : Delay vs Duty cycle (OUT1)

At very low duty cycles (e.g., 5%), OUT1 shows a **higher propagation delay**, reaching approximately **580 ns**. As the duty cycle increases, the delay **gradually decreases** and reaches a **stable plateau** between 20% and 80%, where it remains in the range of **535–545 ns**.

Beyond **85% duty cycle**, the delay begins to **drop sharply** — reaching values as low as **470 ns**. This instability likely results from **pulse compression effects**, where the remaining low-state interval is too narrow to maintain consistent edge shaping or internal timing.

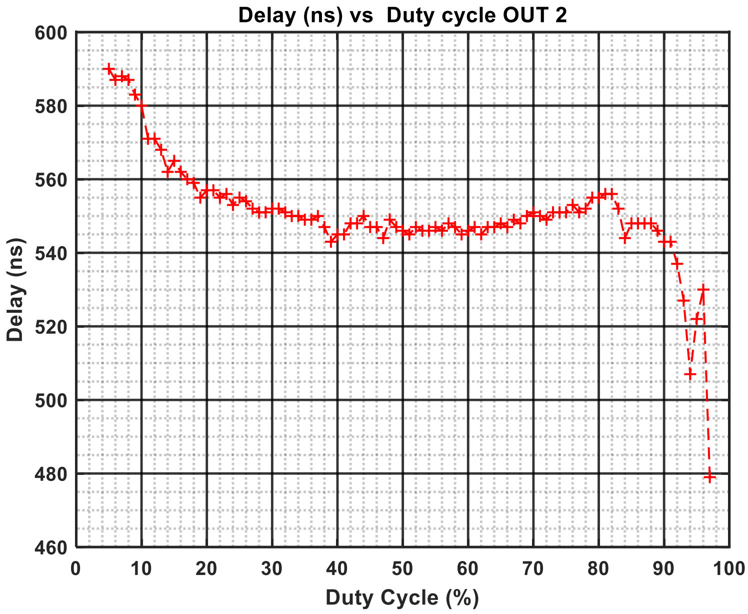


Figure 9 : Delay vs Duty Cycle (OUT2)

OUT2 follows a similar delay profile as OUT1, showing **high delay** at low duty cycles, a **stable midrange plateau**, and a **sharp drop-off** beyond 90%. The delay values for OUT2 are slightly higher (typically by ~5 ns), which may be due to **minor internal asymmetries** in signal routing or logic delays

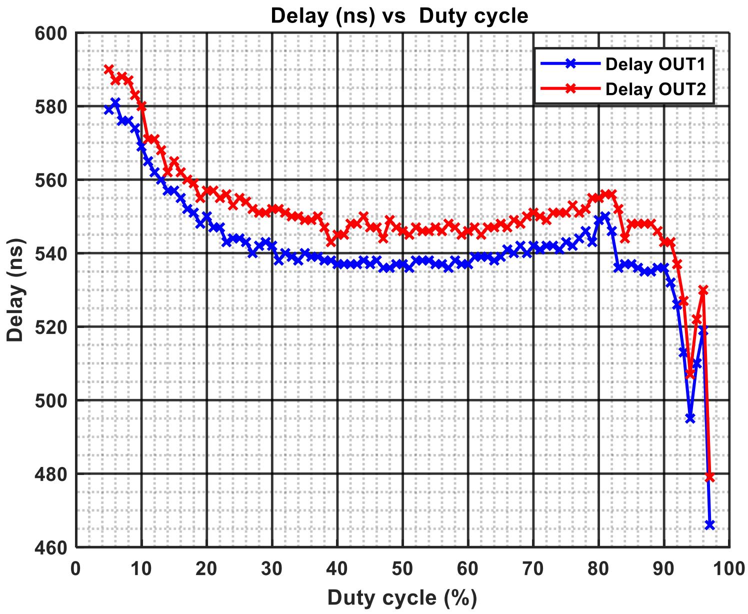


Figure 10 : Delay vs Duty Cycle (OUT1 and OUT2)

The comparison between OUT1 and OUT2 confirms that both outputs behave **symmetrically** across the duty cycle range. The delay mismatch remains below **10 ns** in the stable region (15%–85%), which is acceptable for most full-bridge applications. However, the variation increases noticeably beyond 85%, introducing potential **current asymmetry** in differential drive modes.

Conclusion – Test 3: Delay vs Duty Cycle

- The MP6551 exhibits **nonlinear delay variation** at the **extremes of the duty cycle range**.
- The most **stable delay range** is between **15% and 80% duty cycle**, where the propagation delay remains tightly controlled (~535–545 ns).
- At **very low (<10%)** or **very high (>90%)** duty cycles, the delay becomes **unstable**, which may impact ADC synchronization and current measurement accuracy.

For precise control, it is recommended to either:

- **Clamp duty cycles in firmware** to remain within the linear region (15%–80%), or
- **Apply lookup-table-based delay compensation** when operating outside this range.